

C_2 -Equivariant Spherical Realizers for Pythagorean Vertex Counts

An expository note

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Abstract

For a Pythagorean triple $(a, b, c) = (m^2 - n^2, 2mn, m^2 + n^2)$ we describe explicit C_2 -equivariant finite sets on S^2 and S^3 whose convex hulls are full-dimensional and have exactly

$$a + 2b + c = 2m(m + 2n) \quad \text{and} \quad a + 2b + 2c + e = (m + n)(3m + n) + e$$

vertices, respectively, where $e \geq 1$ is a freely declared integer top layer. The involution fixes the singly counted layers setwise and exchanges the doubly counted layers in mirror pairs, so the coefficient pattern $(1, 2, 1)$, respectively $(1, 2, 2, 1)$, is realized as equivariant layer data. *This note is expository and claims no new theorem:* the vertex-count sequences and the corridor identity are recorded in the OEIS, the underlying convex-geometric fact (finite subsets of spheres are in convex position) is textbook material, and the arithmetic identities are one line from the classical perimeter formula. The purpose of the note is to present the mechanism cleanly, to fix its arithmetic context, to record a small computed boundary atlas of the corridor realizers, and to state the correct integer form of a four-dimensional count for which an irrational “formula” once circulated in a withdrawn precursor manuscript.

Keywords: Pythagorean triples; convex polytopes; exposed vertices; spherical point sets; equivariant configurations.

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1 Introduction

The Pythagorean triples obtained from consecutive Euclidean parameters $m = k + 1$, $n = k$ have the form

$$(a_k, b_k, c_k) = (2k + 1, 2k(k + 1), 2k(k + 1) + 1), \quad k \geq 1.$$

This is the classical family attributed to Pythagoras: the odd numbers as first legs, with hypotenuse exceeding the even leg by one. It is thoroughly documented; in the OEIS the three coordinates are A005408, A046092 and A001844, and the entries record the family explicitly [1, 2]. The middle term $b_k = 2k(k + 1)$ is twice the pronic number $k(k + 1)$.

This note studies the vertex-count expression

$$a + 2b + c$$

on this corridor and, more generally, on the full Euclidean parametrization. The construction below should be read as a *realizer* statement: an explicit spherical point configuration whose

convex hull has the prescribed number of vertices, with the coefficient pattern $(1, 2, 1)$ realized as equivariant layer data — the term $2b$ is represented by a two-sheet layer exchanged by a C_2 symmetry, while the a and c layers are fixed setwise. A second construction, one dimension up, realizes the pattern $(1, 2, 2, 1)$ on S^3 with an integer top layer.

What this note is and is not. Everything here is elementary, and the individual ingredients are classical or already recorded:

- the count sequence $a_k + 2b_k + c_k = 2(k+1)(3k+1)$ is twice the octagonal numbers, OEIS **A139267** (our corridor value at k is $A139267(k+1)$);
- the corridor identity itself appears as a 2013 comment of J. M. Bergot on the octagonal entry **A000567**: with Euclid parameters $(n, n-1)$ one has $\text{oct}(n) = B + (A + C)/2$, i.e. $A + 2B + C = 2 \cdot \text{oct}(n)$;
- the convex-geometric input — distinct points on a sphere are exposed vertices of their convex hull — is a textbook fact about strictly convex bodies (Lemma 4.5 below, with its one-line proof);
- convex hulls of group orbits are the subject of a far more general theory (orbit polytopes; see [4] and the references there).

Accordingly, *no novelty is claimed*. The note has four modest goals: (i) to record the equivariant two-sheet mechanism behind the coefficient 2 in a self-contained way; (ii) to fix the arithmetic identities $a + 2b + c = 2m(m + 2n)$ and $a + 2b + 2c = (m + n)(3m + n)$ over the full Euclidean parametrization, with their OEIS context and offsets; (iii) to state the correct, integer-valued four-dimensional count; and (iv) to record a small computed boundary atlas of the corridor realizers (Section 8), as data. On the last point: a withdrawn precursor manuscript asserted a four-dimensional “vertex formula” $a + 2b + 2c + c\sqrt{2}$; a vertex count is a cardinality, so no irrational term can occur in it. Section 7 replaces that non-statement by the correct form $a + 2b + 2c + e$ with a declared integer $e \geq 1$.

Scope. The theorems prove existence of full-dimensional spherical realizers, exact vertex counts, and a C_2 two-sheet interpretation of the doubled coefficients. They do not prove uniqueness, an f -vector formula, a classification of polyhedra, or any interpretation beyond convex geometry.

2 Notation

Symbol	Meaning
S^{d-1}	Unit sphere in $\mathbb{R}^d$
$R(n, z, \varphi)$	Regular latitude layer in S^2 : n points, height z , phase φ
σ	Reflection $(x, y, z) \mapsto (x, -y, z)$ of $\mathbb{R}^3$
A, B_+, B_-, C	Stratified layers of the spherical point set
G	Disjoint union $A \sqcup B_+ \sqcup B_- \sqcup C$
P	Convex hull $\text{Conv}(G)$
(a_k, b_k, c_k)	Consecutive-parameter corridor triple
$\text{Vert}(P)$	Vertex set of a convex polytope P
$R_4(n, \zeta, w, \varphi)$	Latitude circle layer in S^3 (Section 7)
σ_4	Reflection $(x, y, z, w) \mapsto (x, y, z, -w)$ of $\mathbb{R}^4$

Phases are read modulo 1 throughout.

3 Regular latitude layers

Definition 3.1. For an integer $n \geq 1$, a height $z \in (-1, 1)$, and a phase $\varphi \in \mathbb{R}$, define

$$R(n, z, \varphi) = \left\{ \left(\sqrt{1 - z^2} \cos 2\pi \left(\frac{i}{n} + \varphi \right), \sqrt{1 - z^2} \sin 2\pi \left(\frac{i}{n} + \varphi \right), z \right) : i = 0, \dots, n-1 \right\}.$$

Each point of $R(n, z, \varphi)$ lies on S^2 .

Lemma 3.2 (Cardinality). *For every $n \geq 1$, $z \in (-1, 1)$, and $\varphi \in \mathbb{R}$,*

$$|R(n, z, \varphi)| = n.$$

Proof. Since $z \in (-1, 1)$, the latitude radius $\sqrt{1 - z^2}$ is nonzero. If two listed points coincide, their angular coordinates differ by an integer multiple of 2π , so $(i - j)/n$ is an integer. Since $0 \leq i, j \leq n - 1$, this forces $i = j$. $\square$

Lemma 3.3 (Phase reversal). *Let $\sigma(x, y, z) = (x, -y, z)$. Then*

$$\sigma(R(n, z, \varphi)) = R(n, z, -\varphi).$$

Proof. The reflection σ sends an angular coordinate θ to $-\theta$. Thus the point indexed by i in $R(n, z, \varphi)$ is sent to the phase class $-i/n - \varphi$, which is one of the phase classes of $R(n, z, -\varphi)$ after reindexing $j \equiv -i \pmod{n}$. $\square$

4 The general C_2 -layer realizer on S^2

Definition 4.1. A C_2 -layer realizer of type (a, b, c) is a finite set

$$G = A \sqcup B_+ \sqcup B_- \sqcup C \subset S^2$$

together with an involutive isometry σ of S^2 such that

$$|A| = a, \quad |B_+| = |B_-| = b, \quad |C| = c,$$

and

$$\sigma(A) = A, \quad \sigma(C) = C, \quad \sigma(B_+) = B_-.$$

If every point of G is a vertex of $\text{Conv}(G)$, we call it a *vertex realizer*.

Let a, b, c be positive integers and let $h \in (0, 1)$. Define

$$A = R(a, h, 0), \quad B_+ = R\left(b, 0, \frac{1}{4b}\right), \quad B_- = R\left(b, 0, -\frac{1}{4b}\right), \quad C = R(c, -h, 0),$$

and set $G = A \cup B_+ \cup B_- \cup C$.

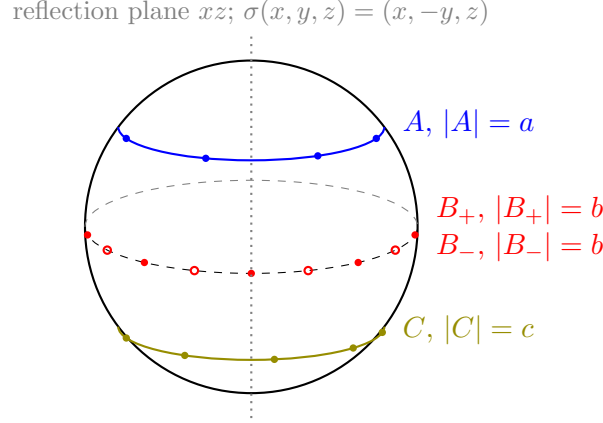


Figure 1: Schematic C_2 -layer realizer of type (a, b, c) . The reflection σ fixes the A - and C -layers setwise and exchanges the two alternating equatorial sheets B_+ (filled) and B_- (open).

Lemma 4.2 (Layer disjointness). *The four layers A , B_+ , B_- , and C are pairwise disjoint.*

Proof. The layers A , $B_+ \cup B_-$, and C have distinct z -coordinates h , 0 , and $-h$, so only B_+ and B_- require attention. If a point of B_+ equaled a point of B_- , then for some integers $0 \leq i, j \leq b-1$,

$$\frac{i}{b} + \frac{1}{4b} \equiv \frac{j}{b} - \frac{1}{4b} \pmod{1},$$

equivalently $i - j + \frac{1}{2} \in b\mathbb{Z}$, which is impossible because the left side is a non-integer half-integer while $b\mathbb{Z} \subseteq \mathbb{Z}$. $\square$

Lemma 4.3 (Equivariance). *The reflection $\sigma(x, y, z) = (x, -y, z)$ satisfies*

$$\sigma(A) = A, \quad \sigma(C) = C, \quad \sigma(B_+) = B_-, \quad \sigma(B_-) = B_+.$$

Proof. This is phase reversal (Lemma 3.3) applied to the phases 0 , $\frac{1}{4b}$, and $-\frac{1}{4b}$. $\square$

Lemma 4.4 (Alternating equatorial layer). *The union $B_+ \cup B_-$ is the regular equatorial $2b$ -gon $R(2b, 0, \frac{1}{4b})$ with its vertices divided into two alternating classes.*

Proof. The phases in B_+ are $\frac{i}{b} + \frac{1}{4b} = \frac{4i+1}{4b}$, $i = 0, \dots, b-1$: the odd classes congruent to 1 modulo 4. The phases in B_- are $\frac{i}{b} - \frac{1}{4b} = \frac{4i-1}{4b}$; for $i = 1, \dots, b-1$ this equals $\frac{4(i-1)+3}{4b}$, and for $i = 0$ it is congruent modulo 1 to $\frac{4b-1}{4b} = \frac{4(b-1)+3}{4b}$. Hence B_- gives the odd classes congruent to 3 modulo 4. Together the two sets give all phases $\frac{2\ell+1}{4b}$, $\ell = 0, \dots, 2b-1$, which are exactly the phases of $R(2b, 0, \frac{1}{4b})$; the two sheets are the two alternating classes. $\square$

Lemma 4.5 (Exposed vertices on the sphere). *Let $X \subset S^{d-1}$ be finite with distinct points. Then every point of X is an exposed vertex of $\text{Conv}(X)$.*

Proof. Fix $p \in X$ and consider the linear functional $\ell_p(x) = p \cdot x$. Then $\ell_p(p) = 1$. For any $q \in X$ with $q \neq p$, the Cauchy–Schwarz inequality gives $p \cdot q < 1$, because p and q are distinct unit vectors. Thus ℓ_p has a unique maximum at p on X . If x is a convex combination of points of X , then $\ell_p(x)$ is the same convex combination of the values $\ell_p(q)$; since all values except $\ell_p(p)$ are strictly less than 1, equality occurs only at $x = p$. Hence p is exposed in $\text{Conv}(X)$. $\square$

This is the standard strict-convexity fact (see e.g. [3] for general polytope background); it is stated with proof only to keep the note self-contained.

Proposition 4.6 (General C_2 vertex count). *The set G above satisfies*

$$|\text{Vert}(\text{Conv}(G))| = a + 2b + c.$$

Proof. By layer disjointness (Lemma 4.2) and cardinality (Lemma 3.2), G has $a + b + b + c$ distinct points, all on S^2 . By the exposed-vertex lemma every point is an exposed vertex of $\text{Conv}(G)$. $\square$

Remark 4.7. The proposition is an existence lemma, not a canonical construction for arbitrary triples. Its role is to isolate the equivariant mechanism by which the coefficient pattern $(1, 2, 1)$ is represented: the middle layer is realized as two alternating sheets exchanged by C_2 . The Pythagorean corridor enters only after this mechanism is fixed.

5 Pythagorean counts and their arithmetic context

For integers $m > n \geq 1$, the Euclidean parametrization

$$a = m^2 - n^2, \quad b = 2mn, \quad c = m^2 + n^2$$

produces Pythagorean triples, and every primitive triple arises (up to order of the legs) from coprime m, n of opposite parity [2]. The corridor of the introduction is the slice $(m, n) = (k+1, k)$. Recall the classical perimeter formula

$$a + b + c = 2m(m + n).$$

Lemma 5.1 (Closed corridor count). *For every $k \geq 1$,*

$$a_k + 2b_k + c_k = 2(k+1)(3k+1).$$

Proof. $a_k + 2b_k + c_k = (2k+1) + 4k(k+1) + (2k(k+1)+1) = 6k^2 + 8k + 2 = 2(k+1)(3k+1)$. $\square$

Lemma 5.2 (General count identities). *For all $m > n \geq 1$, with (a, b, c) as above,*

$$a + 2b + c = 2m(m + 2n), \quad a + 2b + 2c = (m + n)(3m + n).$$

Proof. $a + 2b + c = (m^2 - n^2) + 4mn + (m^2 + n^2) = 2m^2 + 4mn = 2m(m + 2n)$, and $a + 2b + 2c = (m^2 - n^2) + 4mn + 2(m^2 + n^2) = 3m^2 + 4mn + n^2 = (m + n)(3m + n)$. Equivalently: the first is the perimeter plus b , the second the perimeter plus $b + c$. $\square$

Remark 5.3 (OEIS context; nothing here is new). The corridor slice of the first identity gives $2(k+1)(3k+1)$, which is twice the octagonal number $(k+1)(3k+1)$: the sequence 16, 42, 80, 130, ... is OEIS **A139267** (“twice octagonal numbers $2N(3N-2)$ ”), our value at k sitting at index $N = k+1$. The identity itself, in the equivalent form $\text{oct}(N) = B + (A + C)/2$ for Euclid parameters $(N, N-1)$, is a 2013 comment of J. M. Bergot on **A000567**. The corridor slice of the second identity gives $(2k+1)(4k+3)$: the sequence 21, 55, 105, ... sits in OEIS **A033567** ($a(N) = (2N-1)(4N-1)$), again at index $N = k+1$; note that $A033567(0) = 1$ and $A033567(1) = 3$ are *not* corridor values. The corridor family itself — odd first legs, hypotenuse exceeding the even leg by one — is explicitly documented in the entries **A005408**, **A046092**, **A001844** [1].

6 Realizer theorems on S^2

Theorem 6.1 (Corridor realizer theorem). *Let $k \geq 1$ and $h \in (0, 1)$. Define*

$$A_k = R(a_k, h, 0), \quad B_{k+} = R\left(b_k, 0, \frac{1}{4b_k}\right), \quad B_{k-} = R\left(b_k, 0, -\frac{1}{4b_k}\right), \quad C_k = R(c_k, -h, 0),$$

and $G_k = A_k \cup B_{k+} \cup B_{k-} \cup C_k$. Then $\sigma(A_k) = A_k$, $\sigma(C_k) = C_k$, $\sigma(B_{k\pm}) = B_{k\mp}$; every point of G_k is an exposed vertex of $P_k = \text{Conv}(G_k)$; and

$$|\text{Vert}(P_k)| = |G_k| = a_k + 2b_k + c_k = 2(k+1)(3k+1).$$

Moreover P_k is full-dimensional in $\mathbb{R}^3$.

Proof. Equivariance and the vertex count follow from Proposition 4.6 and Lemma 5.1. For full-dimensionality: $a_k \geq 3$, so A_k contains three non-collinear points in the plane $z = h$; since C_k is nonempty and lies in the distinct plane $z = -h$, the affine span of G_k has dimension three. $\square$

Corollary 6.2 (Full Euclidean parametrization). *Let $m > n \geq 1$, $(a, b, c) = (m^2 - n^2, 2mn, m^2 + n^2)$, and $h \in (0, 1)$. The same construction with layer sizes (a, b, b, c) yields a C_2 -equivariant vertex realizer with*

$$|\text{Vert}(P)| = a + 2b + c = 2m(m + 2n),$$

full-dimensional in $\mathbb{R}^3$.

Proof. Proposition 4.6 applies to arbitrary positive integers (a, b, c) , and $a = m^2 - n^2 \geq (n+1)^2 - n^2 = 2n+1 \geq 3$ gives full-dimensionality as in Theorem 6.1. The count is Lemma 5.2. $\square$

7 An integer-layer realizer on S^3

We now iterate the mechanism once, one dimension up, for the coefficient pattern $(1, 2, 2, 1)$. A vertex count is a cardinality; consequently *every* layer of any such pattern must have an integer size. The correct four-dimensional statement therefore carries a freely declared integer top layer $e \geq 1$; no irrational multiple of a layer size (such as the $c\sqrt{2}$ of the withdrawn precursor) can occur in a vertex count.

Definition 7.1. For an integer $n \geq 1$, parameters ζ, w with $\zeta^2 + w^2 < 1$, and a phase $\varphi \in \mathbb{R}$, define the *latitude circle layer*

$$R_4(n, \zeta, w, \varphi) = \left\{ \left(\rho \cos 2\pi\left(\frac{i}{n} + \varphi\right), \rho \sin 2\pi\left(\frac{i}{n} + \varphi\right), \zeta, w \right) : i = 0, \dots, n-1 \right\},$$

where $\rho = \sqrt{1 - \zeta^2 - w^2} > 0$. Each point lies on S^3 , and $|R_4(n, \zeta, w, \varphi)| = n$ by the argument of Lemma 3.2. We call (ζ, w) the *signature* of the layer.

Let $\sigma_4(x, y, z, w) = (x, y, z, -w)$. Since σ_4 acts only on the last coordinate,

$$\sigma_4(R_4(n, \zeta, w, \varphi)) = R_4(n, \zeta, -w, \varphi).$$

Fix positive integers a, b, c, e and parameters

$$\zeta_a \neq \zeta_e, \quad t_b, t_c > 0, \quad (\zeta_b, t_b) \neq (\zeta_c, t_c),$$

all subject to $\zeta^2 + w^2 < 1$ for each layer below. Define

$$A = R_4(a, \zeta_a, 0, 0), \quad E = R_4(e, \zeta_e, 0, 0),$$

$$B_{\pm} = R_4(b, \zeta_b, \pm t_b, 0), \quad C_{\pm} = R_4(c, \zeta_c, \pm t_c, 0),$$

and $G = A \cup B_+ \cup B_- \cup C_+ \cup C_- \cup E$.

Lemma 7.2 (Disjointness by signatures). *The six layers are pairwise disjoint.*

Proof. Under the stated parameter conditions the six signatures $(\zeta_a, 0)$, $(\zeta_e, 0)$, $(\zeta_b, \pm t_b)$, $(\zeta_c, \pm t_c)$ are pairwise distinct: the two fixed layers differ in ζ ; each mirror pair differs in the sign of $w \neq 0$; the same-sign B/C pairs differ because $(\zeta_b, t_b) \neq (\zeta_c, t_c)$; the mixed-sign B/C pairs differ in the sign of w , since $t_b > 0 > -t_c$ and $-t_b < 0 < t_c$; and fixed layers differ from mirror layers because $t_b, t_c > 0$. Points in layers with distinct signatures differ in their last two coordinates. $\square$

Remark 7.3. On S^2 the two B -sheets shared the same latitude and the disjointness argument needed the half-integer phase computation of Lemma 4.2; on S^3 each layer carries its own signature, so no phase argument is needed. The price is that the mirror pairs are parallel n -gons rather than interleaved sheets of a single $2n$ -gon.

Theorem 7.4 (Integer-layer realizer on S^3). *With the data above, σ_4 fixes A and E setwise (indeed pointwise) and exchanges $B_+ \leftrightarrow B_-$ and $C_+ \leftrightarrow C_-$; every point of G is an exposed vertex of $P = \text{Conv}(G)$; and*

$$|\text{Vert}(P)| = a + 2b + 2c + e.$$

Moreover P is full-dimensional in $\mathbb{R}^4$ whenever $a \geq 3$. In particular, for a Pythagorean triple from Euclid parameters $m > n \geq 1$ and any declared integer $e \geq 1$,

$$|\text{Vert}(P)| = (m + n)(3m + n) + e.$$

Proof. Equivariance is the displayed action of σ_4 on signatures. Disjointness is Lemma 7.2, so $|G| = a + 2b + 2c + e$ with all points distinct on S^3 ; the exposed-vertex lemma (Lemma 4.5, stated for S^{d-1}) gives the vertex count. For full-dimensionality: A contains three non-collinear points in the 2-plane $\{z = \zeta_a, w = 0\}$; a point of E (with $z = \zeta_e \neq \zeta_a, w = 0$) raises the affine dimension to three inside the hyperplane $w = 0$; a point of B_+ (with $w = t_b \neq 0$) raises it to four. The Pythagorean count is Lemma 5.2, and $a = m^2 - n^2 \geq 3$ always holds. $\square$

Remark 7.5 (A symmetric choice of top layer). The choice $e = a$ gives

$$a + 2b + 2c + a = 2(a + b + c) = 4m(m + n),$$

twice the perimeter — the pattern $(1, 2, 2, 1)$ becomes the “doubled” pattern $2(1, 1, 1)$ on the triple itself.

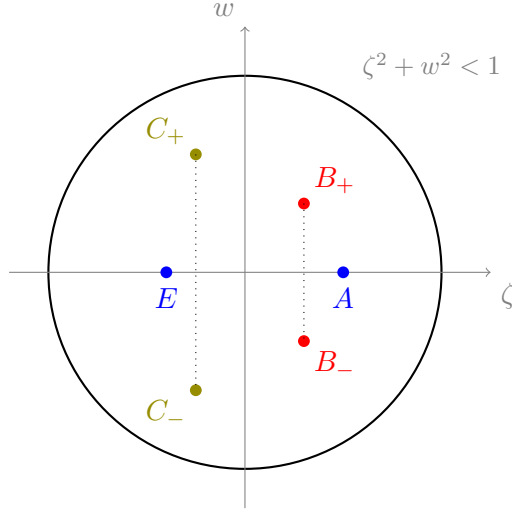


Figure 2: Signature diagram of the S^3 construction: each latitude circle layer is a point in the open unit disk of the (ζ, w) -plane. The reflection σ_4 acts as $w \mapsto -w$: the fixed layers A, E sit on the axis, the doubled layers form mirror pairs.

8 A small boundary atlas

The theorems above concern vertex counts only; edge and face numbers are *not* determined by them and belong to the specific realizer. This section records, purely descriptively, the computed boundary data of the corridor realizers $P_k = \text{Conv}(G_k)$ of Theorem 6.1. The computation takes the exact construction (phases $0, \pm \frac{1}{4b_k}, 0$), merges the triangulated hull facets into true planar faces, counts $E = \frac{1}{2} \sum_{\text{faces}} (\text{face size})$, and verifies Euler's relation $V - E + F = 2$ as a consistency check. For every $k = 1, \dots, 20$ and every tested height $h \in \{0.2, 0.5, 0.8\}$ the resulting (V, E, F) and face-size profile were identical across the three heights; all 60 runs pass the vertex-count and Euler checks.

k	(a_k, b_k, c_k)	V	E	F	face profile
1	(3, 4, 5)	16	38	24	21 triangles, 2 quads, c -gon cap
2	(5, 12, 13)	42	106	66	62 triangles, 2 quads, a -, c -gon caps
3	(7, 24, 25)	80	206	128	124 triangles, 2 quads, caps
4	(9, 40, 41)	130	338	210	206 triangles, 2 quads, caps
5	(11, 60, 61)	192	502	312	308 triangles, 2 quads, caps
6	(13, 84, 85)	266	698	434	430 triangles, 2 quads, caps
7	(15, 112, 113)	352	926	576	572 triangles, 2 quads, caps
8	(17, 144, 145)	450	1186	738	734 triangles, 2 quads, caps
9	(19, 180, 181)	560	1478	920	916 triangles, 2 quads, caps
10	(21, 220, 221)	682	1802	1122	1118 triangles, 2 quads, caps

The structure visible in the profile column is uniform over the computed range: the plane $z = h$ supports the regular a_k -gon of A_k as a top cap and $z = -h$ supports the c_k -gon of C_k as a bottom cap (for $k = 1$ the top cap is a triangle and is counted among the triangles); exactly two quadrilateral faces occur, sitting astride the reflection plane — note that any two σ -paired

points span translation vectors parallel to the y -axis, so two mirror pairs are always coplanar, and at the two seams of the configuration such a quadruple is extremal in the computed range; all remaining faces are triangles.

Remark 8.1 (No claim). Everything in this section is computed data for the specific realizers at the tested parameters ($k \leq 20$, three heights). For the record, for all computed k ($1 \leq k \leq 20$, all three heights) the data fit $E = 16k^2 + 20k + 2$ and $F = 10k^2 + 12k + 2$ (consistent with Euler and $V = 6k^2 + 8k + 2$). We make *no* claim that the face structure, the fits, or the h -independence persist beyond the computed range, and we do not pursue proofs; f -vectors of the general construction are outside the scope fixed in Section 1.

9 Context, prior art, and terminology

Prior art. The arithmetic layer of this note is known or elementary: A139267 (count sequence), the Bergot comment on A000567 (corridor identity), A001844/A046092/A005408 (the family), and the perimeter formula from which Lemma 5.2 is one line. The geometric layer rests on the textbook exposedness fact (Lemma 4.5); realizing *any* prescribed number of hull vertices by points on a sphere is therefore trivial as pure existence. The residual content of the constructions is only the explicit equivariant layer bookkeeping. Convex hulls of orbits of finite groups are studied in much greater generality in the orbit-polytope literature [4, 5]; nothing in this note adds to that theory.

Terminology. “Realization” has an established, different meaning in rigidity theory (embeddings of graphs on spheres, e.g. counting realizations of Laman graphs [6]). Here *realizer* means only: a configuration whose convex hull attains a prescribed vertex count with prescribed equivariant layer structure.

10 Remarks and limitations

Different choices of the parameters (h ; the signatures; the phases) yield non-congruent realizers with the same vertex count, so the results are equivariant realizability statements, not uniqueness statements. The constructions give vertex realizers, not f -vector theorems: edge and face counts are not determined here. The general propositions are existence lemmas; they are not canonical or natural in any categorical sense. And, as stated throughout, none of the mathematical content is new; the note is a consolidation with corrected statements and fixed arithmetic context.

A Computational reproducibility

The proofs above are independent of computation; the following implementation checks accompany the project.

Script	Cases	Counts	Checks	Result
<code>verify_c2_realizer.py</code>	$k = 1, \dots, 20$	16 – 2562	hull count = $2(k+1)(3k+1)$	20/20
<code>verify_s3_realizer.py</code>	50 runs (48 distinct)	22 – 1300	hull count, disjointness, σ_4	50/50
<code>check_oeis_prior_art.py</code>	11 checks	—	identities, OEIS terms, offsets	11/11
<code>atlas_boundary.py</code>	$k \leq 20$, 3 heights	—	merged f -vectors, Euler	60/60

The S^2 script rebuilds G_k and checks $|\text{Vert}(\text{Conv}(G_k))| = 2(k+1)(3k+1)$ for $k = 1, \dots, 20$ (`results/baseline_k1_20.csv`). The S^3 script checks Theorem 7.4 on the corridor $k = 1, \dots, 12$ with $e \in \{1, a_k, 7\}$ and on six further non-corridor Euclid pairs with $e \in \{1, 5\}$ (50 runs, 48 distinct cases; two duplicates arise at $(m, n) = (4, 3)$, where $a_k = 7$ and the pair also appears in the general list), verifying layer disjointness, σ_4 -equivariance, and the four-dimensional hull vertex count (`results/s3_realizer_check.csv`). The OEIS script re-derives the identities of Lemma 5.2 symbolically and diffs the archived OEIS entries against the computed baselines (`results/oeis_prior_art_check.txt`). The atlas script produces the merged-face data of Section 8 (`results/atlas_boundary.csv`). These computations are implementation checks and descriptive data, not ingredients in the proofs.

References

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